Unit 1:

The concept of cyber security is online protection and protection for infrastructure from hackers who will use someone's information for their personal benefit. For example, a hacker got a hold of someone's bank info and is now withdrawing cash maxing credit cards etc. So, to prevent this from happening company's spend millions of £ on security in hopes of stopping data breaches they also spend millions of £ on it because of the Data Protection Act of 2018 makes it so they can be held responsible for it being stolen and make companies transparent on what its being used for and how it's being used.

Identity- Is someone who has verified access to sensitive information like someone bank info, address, ID/ Passport etc but doesn't do anything with it because he

Confidentiality- is to ensure that the data is protected/ secure

Integrity- means that a cyber security specialist it responsible for ensuring that websites are trustworthy and has accurate information. They also have a responsibility to ensure there are appropriate measures in place to prevent alteration of documents website etc.

Availability- This means that personal information and other information like what you could find on a website and must always be available. This relies on well-maintained handwear, software, and reliable internet.

Threat- a cyber security threat refers to an act the seeks to damage a website a person online life. Threats include ransomware, malware, DOS/DDOS, account takeover, SQL Injection, Cross-scripting.

Vulnerability- is a weakness that can be exploited to gain northernised access to a system this allows that person to install malicious software/ take over a computer, network, or a server

Risk- in cyber security refers to the potential exposure of a company system and information. A type of risk is malware however that's just one risk other risks include theft of intellectual property, productivity loss and others.

Hazard- Is anything that could lead to a data breach or data loss. Hazards can also include the types of companies. You could have the most cyber secure building on the plaint but if your contract companies have mediocre cyber security then they become a backdoor for cyber criminals.

Unit 1 - Q4 -Define core terminology used in cyber security

Malicious Software is something that has being made by a computer person to exploit a loophole in a computer system in hopes of destroying or steeling their victims' belongings.

distributed denial of service (DDoS) is

cloud is something that stores people's information on a server in a Wherehouse anywhere on the planet and can be accessed at any time by the person who owns it.

software is a few thousand lines of code that make you able to use your device however you like for example, navigation buttons, read, tell the time/ day/ month/ year all without needing to use/ understand binary or any other coding language.

Domain is an internet address or name, a domain/ domain name is the location of a website.

Exploit is a peace of code that takes advantage of venerability

Breach is where unorthernized persons gain access to data to steal, expose or release it.

Firewall a security system that monitors and controls the network traffic.

Encryption is scramble what the original message was with symbols, numbers, different letter, and only those who have the encryption key will see the actual message.

Bring Your Own Device (BYOD) is where people bring their own laptops, tablets or other handheld device to work so they can access work files on their own device and the company doesn't have to issues a device to every single person who works there.

Penetration testing (pen testing) is simulating a cyber-attack on a company network or appucations to identify vulnerabilities.
DDoS Stands for Distributed Denial- of-Service a cyber-attack in which a network or website are flooded with excessive traffic to disrupt normal access.

Unit 1 - Q5 -Explain the terms good actors and bad actors

A good and bad actor

In cyber security an actor is a person or group that carries out actions involving computer systems, network, or data. Actors can be described as good or bad depending on their intentions, or if they have been authorized to carry out an attack.

Bad Actor

Insider-Threat Ex-employee a former employee who uses information accounts or accesses they previously had to cause harm to a company.

Black Hat

Is a cybercriminal who deliberately uses hacking knowledge to gain northernised access to computer systems/ networks to steel data, damage systems, or make money illegally.

Script Kiddies

Is a person with limited knowledge who uses tools, scripts, or software created by other hackers to carry out attacks.

Hacktivist

A person or group that uses hacking to promote a political, social, or ideological.

Good Actors

white hat is an ethical hacker who uses hacking knowledge legally and with permition to find weakness in security systems. Explicit premotions to

Certified Penetration Tester is a trained and certified professional who is authorised to perform Penetration testing. Hired to perform whin his scope and authorisations.

Unit 1 - Q6 - Distinguish typical behaviours of good actors and bad actors

Good Actors – protect and defend systems ethically.

Operate with explicit permission, follow responsible disclosure, Document actions, use control testing methods, Focused on defence and improvement, Adherence to ethical and compliance

Bad Actors carry out unauthorised activities with harmful or malicious intent. Their motivations and methods vary.

Cybercriminals

Motivations: Financial gain, operates like a business, often working in organised groups. Use attacks such as phishing, ransomware and data theft to make money. May sell stolen data, malware or access to compromised systems.

Nation Stake - Fund by their country to sabotage another/ collect intel on them, long term stealth access, highly sophisticated and well resourced

Motivations: National interests, espionage or sabotage. Are often sponsored or supported by government. Target other countries, governments or important organisations to gather intelligence or disrupt operations. May maintain long-term, stealthy access to systems. Often have Signiant funding advanced skills and recourse.

Hacktivists

Politically motivated, seek visibility, Skill differs greatly

Motivations: Political, social or ideological beliefs. Attack systems to promote a cause or draw public attention to an issue. May use website defacement, data leaks or DOs Attacks. Their technical skills and resources vary greatly.

Unit 1 - Q7 - Explain the motivations of good actors and bad actors

Motivations - must include, but are not limited to:

a) bad actor – financial gain, terrorism, political motivation, reputation, disruption, respect, thrill seeking

b) good actor – ethical, job role, improve things, innovation, challenge of being better than a black hat.

Good Actors – protect and defend systems ethically.

Good actors and bad actors are people who use their cyber skills for verry different reasons. Good Actors sometimes called white hat hackers / ethical hackers have permission to test, protect and improve computer systems. Their main motivation is to identify weaknesses before they can be exploited by criminals and to help organisations improve their security. Bad actors on the other hand access or attack systems without permission with the intention of causing harm stealing info to make money or to active another malicious objective.

One motivation for ethical hacker is their ethical responsibility. Ethical hackers want to protect people & organisations and their info from cyber-attacks. They have permission from the organisation and follow agreed upon rules about what they're allowed to test. For example, a company may employ a pen tester to test their website before a cybercriminal can take advantage of any vulnerabilities the tester could find a weakness that allows for unnotarized access and then report it to the organisations so they can get someone in to fix it.

Bad Actors

Bad actors have different motivation. One of the most common is financial gain cyber criminals may treat cybercrime as a business due to the value of the stolen info, compromised accounts and access to computer systems. They may use stuff like malware or ransomware to steel money or demand payment from victims for example, ransomware attackers can gain unauthorised access to organisation systems, encrypt important documents and demand payment in exchange for restoring access.

Unit 1 - Q8 - Identify key sectors that are most vulnerable to a cyber-attack

Key sectors must include, but are not limited to:

a) health

b) manufacturing

c) finance

d) government agencies and departments

e) educational institutions.

There are serval key sectors that are particularly vulnerable to cyber-attacks because they hold vast amounts of personal information and rely heavily on the internet, computers, and networks. Those sectors include Health, Manufacturing, Finance, Government agencies, departments and education institutions, and public transport/ energy grid. Attackers target these sectors for multiple reasons like money, sensitive info, espionage or political reasons.

Healthcare is a particularly valuable target because organizations such as the NHS hold vast amounts of patient info like their medical record, address, bank info, contact info and other information that may be used to target the patient directly. Healthcare or organizations also rely on computer systems to manage appointments, patient/ patient records, communication, medical equipment and other services. Cybercriminals mostly use ransomware against organizations because disrupting access to important systems can put significant pressure on the organizations to get it back up and running. A real-world example is the 2024 Synnovis cyber-attack which affected pathology in London and caused significant disruption to the health service. Another venerability is though their supply changes because hospitals depend on external companies for most of their equipment and its maintenance and other supplies

Manufacturing is another venerable sector because modern factories rely heavy upon computer systems, networks and operational tech. An attacker who compromises these systems could potentially stop production, damage equipment, steel intellectual property. Can therefore be an attractive target for ransomware because the smallest bit of down time results in massive financial loss.

Unit 1 - Q9 - Compare the motivations for a cyber-attack in key sectors

The reasons for cyber-attacks vairy dependent on the sector some want money, some want attention for something, or to cause disruptions.

Sector	**Main motivations**	**Example**
Health	Steal personal info, gain money, Disruption and extortion.	Ransomware attack could be used to steel personal info and to gain money.
Manufacturing	Finance gains, industry espionage, Disruption, and intellectual theft.	An attack could steal information about a company next step or development plans.

Finance	Finance gain, fraud, theft of finance information.	A criminal could attack a bank to steel money or personal information from the clints.
Government	Espionage, political objections, disruptions and theft of sensitive info.	A hostels state could attack governments systems to obtain confidential info.
Education	Finance gain, data theft and disruption.	Attackers could use ransomware to prevent a university from accessing its systems and demand payment.

Although finance gain is common in almost all sectors, the potential information/ disruption to that organisation, Government, etc can be more devastating.

Unit 1 - Q10 - Consider how an actor may carry out a cyber-attack

Please note: You must state the type of attack and how it can be carried out by the actor.

A cyber-attack can be carried out in different ways depending on the actors aims the organisation being, targeted and the vulnerabilities to exploit.

Pushing attack:

Type of attack: Social engineering/phishing.

Cyberattack that uses fraudulent email, text message, phone call, or website to trick people into sharing sensitive data to download malware.

Ransomware

Type of attack: Denial of Service

An attacker can send a verry large number of requests to a website or online service. In a DDoS attack the traffic can come from many compromised devices at once. That target can become overloaded making the service slow.

Unit 1 - Q11 - Describe the term security by design

Security is an approach to software and hardware development that seeks to make security an important part of a system from beginning rather than adding security after the product has already been made. This means that devs consider security risks and vulnerabilities during design stages.

Unit 1 - Q12 - Explore the principles of security by design.

Note:

You could explore the principles set out by organisations such as the National Cyber Security Centre (NCSC) Security by design means considering throughout the design and development of systems instead of treating it as something that's added later. The NCSC provides 5 main principles for design secure systems. Those principles are:

Establish the context before designing a system

The company should understand what the system is used for what info it contains who needs access to it and what risks it faces. The NCSC also recommend understanding the threat model to understand how attacks could exploit it.

Make compromises difficult

Systems should be designed to make it as difficult as possible for an atter to gain northernised access. This includes reducing the attacker's surface validating external data using strong authentication and removing unnecessary features or services.

Make disruption difficult

Systems should be designed to continue working even when they experience an attack or failure. This is especially important for important services where down time could cause serious problems.

Make compromise detection easier

It is important to accept the security controls cannot prevent every possible attack. Systems should therefore be designed so that suspicious activity can be detected quickly.

Reduce the impact

If an atter does successfully gain access the system should be designed to limit how much damage they can cause.

Unit 1 - Q13 - State the consequences of not considering cyber security during the design phase

Note:

You should use the NCSC guidelines to state the consequences of not considering cyber security during the design phase.

Increased vulnerability to attacks systems designed without security considerations are easier for attackers to exploit as they lack robust defence to prevent unauthorized accessed.

Reactive problem-solving addressing security issues after deployment often leads to a reactive approach where vulnerability is patched only after they are exploited.

Higher cost adding security measures post deployment is significantly more expensive than integrating hem during development.

Operational disruptions lack of security can lead to system downtime due to breaches or denial of service attacks.

Unit 1 - Q14 - Evaluate the advantages and disadvantages of security by design

Note: Suggested evidence of assessment could be in report format and could provide a piece of evidence for a learner's portfolio. You could consider an example based on their surroundings (eg swipe card to enter and exit the building).

Advantages

Proactive risk management early detection of vulnerabilities allows for mitigations strategies before development begins.

Cost effective fixing security early is less expensive than addressing them after deployment.

Improved compliance incorporating security measured from the start helps comply with regulations and standers.

Enhance user trust systems designed with security in mind foster user confidence.

Reduce attack risk thoughtful design limits unnecessary features and service that may create vulnerabilities.

Disadvantages of security by design

Higher initial cost security by design can increase the coat of development because companies may need to spend more money on security tools testing specialist staff from the start.

Longer development times adding security measures during the design and development stages can make the development process take longer because more testing is required.

Can be difficult to implement security control can sometimes be complicated to and implement especially when developers do not have enough security knowledge.

Possible impact on usability stronger security measures can sometines make systems more difficult or slower for users to access.